

HW #1: Mixture Density Networks

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Probabilistic Deep Learning — Fall 2026

1 Model and implementation

For an input $\mathbf{x}$ and target $\mathbf{y}$, our mixture density network (MDN) models the conditional density as

$$p(\mathbf{y} | \mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_k(\mathbf{x}), \boldsymbol{\Sigma}_k(\mathbf{x})), \quad \sum_{k=1}^K \pi_k(\mathbf{x}) = 1. \quad (1)$$

The network is trained by minimizing the mean negative log likelihood (NLL). To implement this, we created an NN module that outputs the mixture weights (through a softmax), μ , and Σ (through a softplus, plus an additional term to ensure a minimum scale of $\epsilon > 0$) for each mixture component. We optimize using the log-sum-exp expansion. Our hyperparameters include the number of mixture components K (the number of Gaussians in the mixture), the rank of the covariance matrices (4), learning rate (1e-3), batch size (64), and number of epochs (2000).

2 Synthetic two-dimensional example

The data we used for our 2D synthetic example was generated by sampling 4000 points from two different Gaussian distributions with different means and covariances. Then, we added Gaussian noise to the output. This resulted in a multimodal distribution for the output given the input, see Figure 1. This resulting trained model should show a single Gaussian near the intersection of the distributions, and two separate Gaussians that are increasingly far apart as the input moves away from the intersection (if the MDN learns well). Figure 2 does indeed show this behavior. The MDN adequately finds each mode with respect to the data.

Additionally, we tried a range of values for K , see Figure 3, and found that there was no significant improvement for any K greater than 2. This is expected because the data comes from a simple two-modal distribution.

We also compare these K models to a deterministic regressor. This regressor performed similarly to the $K = 1$ MDN. NLL and RMSE are reported in Table 1, when relevant.

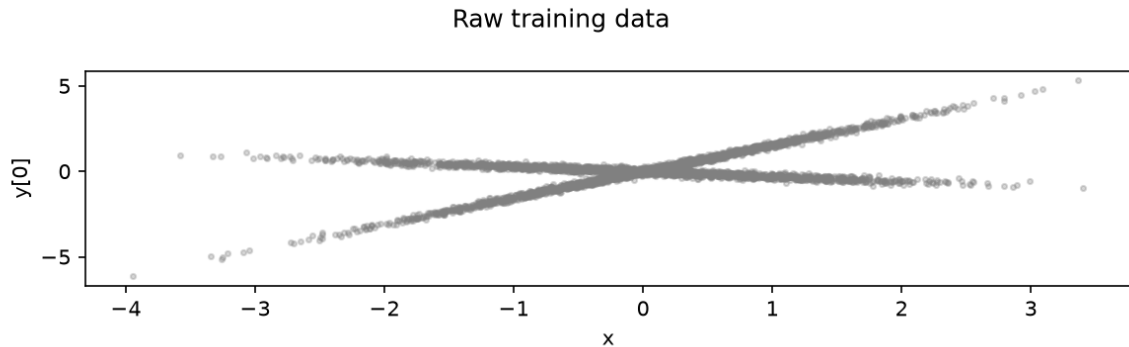


Figure 1: Raw training observations for the synthetic task, showing the multiple output modes associated with the input.

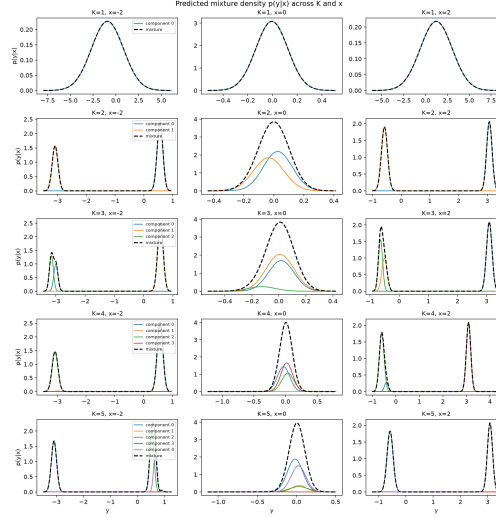


Figure 2: Learned conditional density for the synthetic 2D task over a grid of input and output values. The density should reveal the modes captured by the MDN and how their locations, weights, and widths vary with the input.

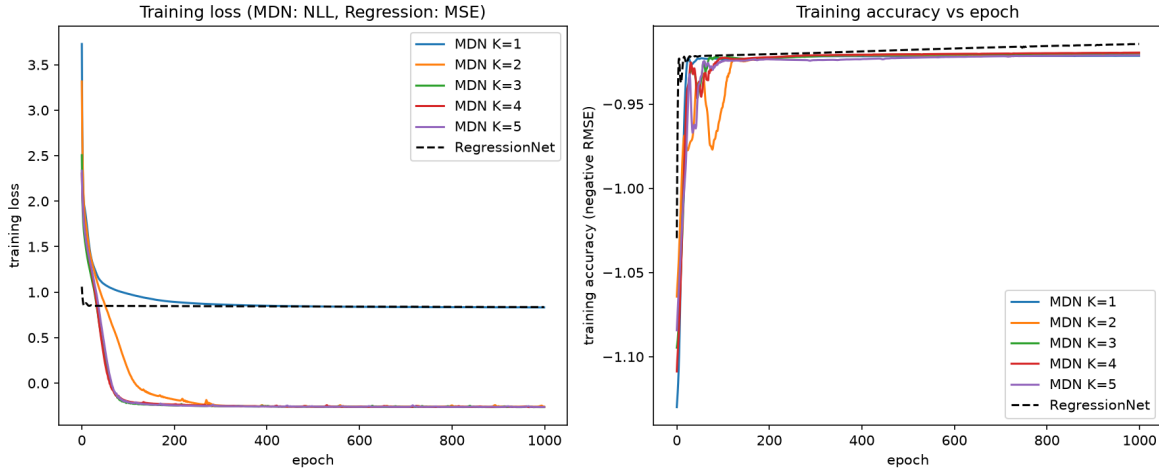


Figure 3: Training objective and RMSE across epochs for the deterministic baseline, single-Gaussian model, and MDNs with different numbers of components.

3 Higher-dimensional realistic data

We use the wind-turbine SCADA dataset as the realistic task. The response, input features, preprocessing, and split are specified below rather than assuming that adjacent time points are independent.

Describe the prediction target and all input dimensions. Summarize dataset size, missing/outlier handling, normalization, and the train/validation/ test protocol. Justify whether the split is random, grouped, or chronological, and explain why multimodal or heteroscedastic output may be plausible.

Table 1: Synthetic-data performance loaded directly from `csvs/synthetic_summary.csv`. NLL is not defined for the deterministic `model`.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	0.914328	0.920263
Single Gaussian	1	0.832287	0.807886	0.921331	0.905627
MDN	2	-0.257680	-0.263095	0.920119	0.907287
MDN	3	-0.263673	-0.265676	0.919707	0.909541
MDN	4	-0.262717	-0.265303	0.919435	0.910978
MDN	5	-0.262428	-0.257620	0.920986	0.905491

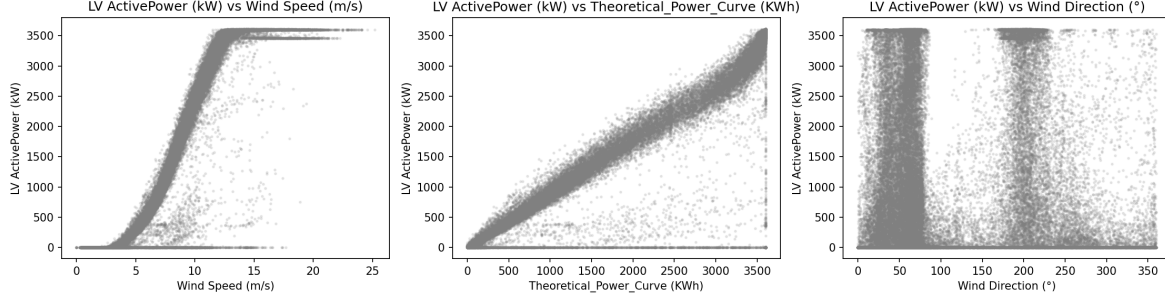


Figure 4: Wind-turbine active power against each available numeric predictor, used to assess multimodality and heteroscedasticity before fitting the models.

Compare training and test behavior on the realistic dataset. Discuss generalization gaps, predictive sharpness and calibration, regions of high uncertainty, the effect of K , and how conclusions differ from the controlled synthetic example. Reference Figures 4 and 5, Figure 6, and Table 2.

4 Conclusion

Summarize when the MDN improves on deterministic and single-Gaussian regression, identify the best-supported mixture count, state limitations, and give one concrete next experiment.

Table 2: Wind-turbine performance loaded directly from `csvs/wind_turbine_summary.csv`. RMSE is reported in kW; NLL is evaluated on standardized targets and is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	383.646149	389.206085
Single Gaussian	1	-0.548006	-0.558157	386.391083	391.403168
MDN	2	-1.100708	-1.081633	391.576721	398.146759
MDN	3	-0.783956	-0.764667	392.832428	399.270935
MDN	4	-1.279749	-1.283014	391.055908	397.277557
MDN	5	-1.281521	-1.287033	392.594330	398.612457

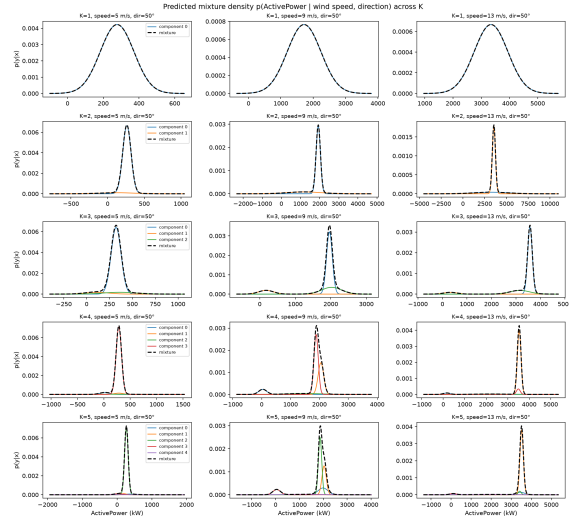


Figure 5: Learned conditional density for the wind-turbine task over a grid of inputs and active-power outputs. Use this view to show heteroscedasticity, multiple operating regimes, and regions where the model assigns diffuse or multimodal probability.

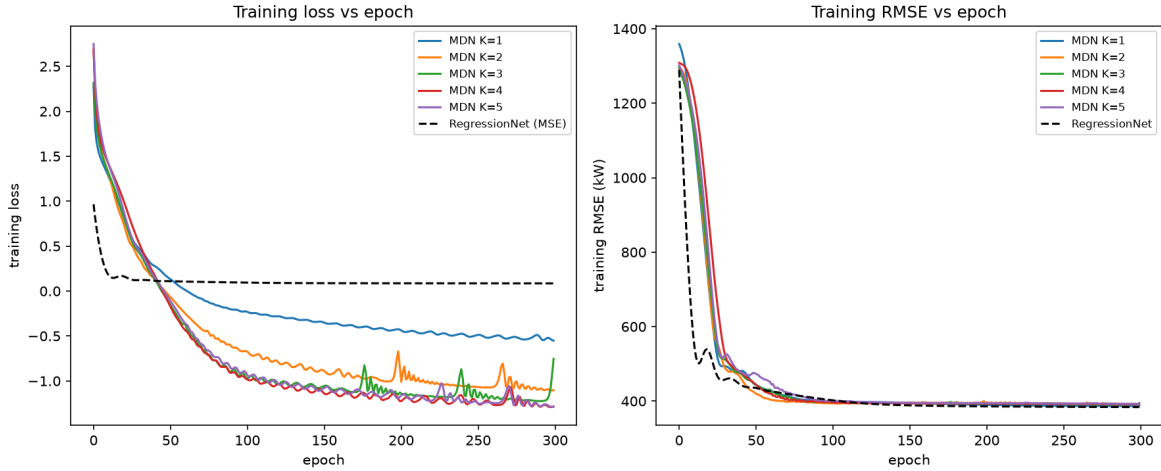


Figure 6: Training and validation/test behavior for the realistic-data models, including the selected checkpoint or stopping point.